Practical 1 :- Swapping With 3rd Variable.

Programiz C Online Compiler

main.c

Run

```c
//Swapping with 3rd Variable.
#include<stdio.h>
int main() {
    int a,b,c;
    printf("Enter First Number:- ");
    scanf("%d",&a);
    printf("Enter Second Number:- ");
    scanf("%d",&b);
    c=a;
    a=b;
    b=c;
    printf("First Number %d \n",a);
    printf("Second Number %d",b);
    return 0;
}
```

Output

```
/tmp/LMrxU1etja.o
Enter First Number:- 6
Enter Second Number:- 8
First Number 8
Second Number 6

=== Code Execution Successful ===
```

Practical 2 :- Swapping Without 3rd Variable.

main.c

Run

```c
//Swapping without 3rd Variable.
#include<stdio.h>
int main() {
    int a,b;
    printf("Enter First Number:- ");
    scanf("%d",&a);
    printf("Enter Second Number:- ");
    scanf("%d",&b);
    a=a-b;
    b=a+b;
    a=b-a;
    printf("First Number %d \n",a);
    printf("Second Number %d",b);
    return 0;
}
```

Output

```
/tmp/VyATrwqpaS.o
Enter First Number:- 7
Enter Second Number:- 5
First Number 5
Second Number 7
```

Practical 3 :- Sum of 5 digit Number without Loop.

main.c

```c
#include <stdio.h>
int main(){
    int ak(int);
    int s=0,x,rem,sum=0;
    printf("Enter a number");
    scanf("%d",&x);
        rem= x%10;
        sum=sum+rem;
        x=x/10;
        rem= x%10;
        sum=sum+rem;
        x=x/10;
        rem= x%10;
        sum=sum+rem;
        x=x/10;
        rem= x%10;
        sum=sum+rem;
        x=x/10;
        rem= x%10;
        sum=sum+rem;
        x=x/10;

    printf("%d",sum);
}
```

Output

```
/tmp/CFu634oRnv.o
Enter a number12345
15
```

Practical 4 :- Sum of 5 digit number using loop.

main.c

```c
#include <stdio.h>
int main(){
    int ak(int);
    int s=0,x,rem,sum=0;
    printf("Enter a number");
    scanf("%d",&x);
    while(x>0)
    {
        rem= x%10;
        sum=sum+rem;
        x=x/10;
    }
    printf("%d",sum);
}
```

Output

```
/tmp/CFu634oRnv.o
Enter a number12345
15
```

Practical 5 :- Reverse of 5 digit number without loop.

```c
#include <stdio.h>
int main(){
    int ak(int);
    int s=0,x,rem,rev=0;
    printf("Enter a number");
    scanf("%d",&x);
        rem= x%10;
        rev=rev*10+rem;
        x=x/10;
        rem= x%10;
        rev=rev*10+rem;
        x=x/10;
        rem= x%10;
        rev=rev*10+rem;
        x=x/10;
        rem= x%10;
        rev=rev*10+rem;
        x=x/10;
        rem= x%10;
        rev=rev*10+rem;
        x=x/10;

    printf("%d",rev);
}
```

Output

```
/tmp/qhoITxSqh0.o
Enter a number12345
54321
```

Practical 6 :- Reverse of 5 digit number with loop.

main.c

```c
#include <stdio.h>
int main(){
    int ak(int);
    int s=0,x,rem,rev=0;
    printf("Enter a number");
    scanf("%d",&x);
    while(x>0)
    {
        rem= x%10;
        rev=rev*10+rem;
        x=x/10;
    }
    printf("%d",rev);
}
```

Output

```
/tmp/QiIOnMhBpF.o
Enter a number12345
54321
```

Practical 7 :- Program to find whether the number is Prime or Not.

main.c

```c
//Program to find whether the number is prime or not.
#include<stdio.h>
int main() {
    int num,n;
    printf("Enter any Number:-");
    scanf("%d",&num);
    n = 2;
    while(n<= num-1)
    {
        if(num%n==0)
        {
            printf("Not Prime");
            break;
        }
        n++;
        }
        if(num==n)
        {
            printf("It is Prime");
        }
        return 0;
    }
```

Output

```
/tmp/ImDFgsNX23.o
Enter any Number:-5
It is Prime
```

Practical 8 :- Program to find out whether the year is leap or not without Function.

main.c

```c
#include<stdio.h>
int main () {
    int year,r;
    int leap(int);
    printf("Enter any Year:-");
    scanf("%d", &year);
    r = leap(year);
    if (r>0)
    {
        printf("It is Leap");
    }
    else
    {
        printf("It is not leap");
    }
    return 0;
    }
    int leap(int y)
    {
        if(y % 4 ==0 && y % 100!=0 || y % 400 ==0)
        {
            return 1;
        }
        else
        {
            return 0;
        }
    }
```

Output

```
/tmp/w7yOn1KSWD.o
Enter any Year:-2005
It is not leap
```

Practical 9 :- Program to find out whether the year is leap or not using function.

main.c

```c
#include<stdio.h>
int main() {
    int year;
    int leap(int);
    printf("Enter any Year:-");
    scanf("%d",&year);
    leap(year);
    return 0;
}
int leap(int y)
{
    if(y % 4 == 0 && y % 100 !=0 || y % 4 == 0)
    {
        printf("The year is Leap");
    }
    else
    {
        printf("Not leap");
    }
}
```

Output

```
/tmp/FWeI7qDCha.o
Enter any Year:-2000
The year is Leap
```

Practical 10 :- Program to find out the Square of float number using Functions.

main.c

```c
#include<stdio.h>
int main() {
    float square(float);
    float n,res;
    printf("Enter any float number:- ");
    scanf("%f",&n);
    res = square(n);
    printf("%f",res);
    return 0;
}
float square(float nn)
{
    float r=1;
    r=nn*nn;
    return r;
}
```

Output

```
/tmp/meEqMbu4la.o
Enter any float number:- 23.11
534.072144
```

Practical 11:- Swap using Call by Address.

main.c

```c
#include <stdio.h>

int main() {
    int x, y;
    void swap(int*, int*);
    printf("x: ");
    scanf("%d", &x);
    printf("y: ");
    scanf("%d", &y);

    swap(&x, &y);
    printf("x=%d\n", x);
    printf("y=%d", y);
}

void swap(int *xx, int *yy) {
    int t;
    t = *xx;
    *xx = *yy;
    *yy = t;
}
```

Output

```
/tmp/vaOIhHWWRD.o
x: 6
y: 8
x=8
y=6
```

Practical 12 :- Performing Arithmetic Operation using Call by Address.

main.c

```c
#include<stdio.h>
int main() {
    int prod_div_sum(float,int,float*,float*,float*);
    float V1;
    int V2;
    float prod,div,sum;
    printf("Enter value 1 : ");
    scanf("%f",&V1);
    printf("Enter value 2 : ");
    scanf("%d",&V2);
    prod_div_sum(V1,V2,&prod,&div,&sum);
    printf("%f%f%f",prod,div,sum);
    return 0;
}
int prod_div_sum(float V11,int V22,float*p,float*d,float*s)
{
    *p=V11*V22;
    *d=V11/V22;
    *s=V11+V22;
}
```

Output

```
/tmp/6WtWv66hPg.o
Enter value 1 : 5
Enter value 2 : 6
30.0000000.83333311.000000
```

1. Program to find the Average of Subjects without Function.

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main.c

```c
//Program to find the average of subjects in a class without function.
#include<stdio.h>
int main() {
    int marks[6],i,sum=0;
    float avg;
    printf("Enter the Marks:- ");
    for(i=0;i<=5;i++)
    {
        scanf("%d",&marks[i]);
    }
    for(i=0;i<=5;i++)
    sum=sum+marks[i];
    avg=(float)sum/i;
    printf("The Average is %f",avg);
    return 0;
}
```

Output

```
/tmp/T6ZQIXqQvB.o
Enter the Marks:- 56
78
98
45
34
56
The Average is 61.166668

=== Code Execution Successful ===
```

2. Program to find the Average of Subjects with Function.

main.c

```c
//Average using array with function
#include<stdio.h>

int main() {
    int marks[6],i;
    float average(int*,int);
    float avg;
    printf("Enter Marks of 6 Subjects:- ");
    for(i=0;i<=5;i++)
    {
        scanf("%d",&marks[i]);
    }
    avg=average(&marks[0],i);
    printf("The Average is %f",avg);
    return 0;
}

float average(int*m,int n)
{
    int s=0,i;
    float av;
    for( i=0;i<=n-1;i++)
    {
        s=s+m[i];
    }
    av=(float)s/i;

    return av;
}
```

Output

```
/tmp/WkDp71Ap5B.o
Enter Marks of 6 Subjects:- 56
48
57
72
49
59
The Average is 56.833332
```

3. Program to Swap side by side in Array.

main.c

```c
//swap side by side in array
#include<stdio.h>

int main() {
    int ar[8], i, swa;
    int swap_al(int*, int); //prototype
    printf("Enter No:- ");
    for(i=0; i<=7; i++) {
        scanf("%d", &ar[i]);
    }
    swa = swap_al(&ar[0],i); //calling
    for(int i = 0; i<=7; i++){
        printf("%d", ar[i]);
    }
    return 0;
}

int swap_al(int*a, int n) {
    int i, j=0, t;
    for(i=0; i<=(n-1)/2; i++) {
        t=a[j];
        a[j]=a[j+1];
        a[j+1]=t;
        j=j+2;
    }

    return *a;
}
```

Output

```
/tmp/Ux0ToPc8gq.o
Enter No:-  3
8
4
2
5
7
9
1
83247519
```

4. Program for doing half Swapping in Arrays.

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main.c

```c
#include<stdio.h>
void swap_half(int* a, int n) {
    int i, j = 0, t;
    for (i = 0; i <= (n - 1) / 2; i++) {
        t = a[j];
        a[j] = a[j + 4];
        a[j + 4] = t;
        j++;
    }
}

int main() {
    int arr[8];
    printf("Enter 8 numbers:\n");
    for (int i = 0; i < 8; i++) {
        scanf("%d", &arr[i]);
    }
    swap_half(arr, 8);
    printf("Array after swapping:\n");
    for (int i = 0; i < 8; i++) {
        printf("%d ", arr[i]);
    }
    return 0;
}
```

Output

```
/tmp/HC2ziMlFUR.o
Enter 8 numbers:
4 6 8 9 5 0 3 5
Array after swapping:
5 0 3 5 4 6 8 9
```

5. Program to Reverse the Array using Function.

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main.c

```c
#include <stdio.h>

void reverse_array(int* a, int n) {
    int i, j, t;
    for (i = 0, j = n - 1; i < j; i++, j--) {
        t = a[i];
        a[i] = a[j];
        a[j] = t;
    }
}

int main() {
    int ar[8], i;
    printf("Enter 8 Numbers:- ");
    for (i = 0; i < 8; i++) {
        scanf("%d", &ar[i]);
    }
    reverse_array(&ar[0], 8); // Pass the array and its size
    printf("Reversed array: ");
    for (i = 0; i < 8; i++) {
        printf("%d ", ar[i]);
    }
    printf("\n");
    return 0;
}
```

Output

```
/tmp/2Lc108pFuZ.o
Enter 8 Numbers:- 1 2 3 5 4 6 7 8
Reversed array: 8 7 6 4 5 3 2 1
```

6. Program of Linear Search .

```
//Linear Search Program
#include <stdio.h>

int main() {
  int Lsearch(int*,int,int);
  int i,arr[10],elm,r;
  printf("Enter the Array elements:-
");
  for(i=0;i<=9;i++)
  {
     scanf("%d",&arr[i]);
  }
  printf("Enter the element to be
searched:");
  scanf("%d",&elm);
  r=Lsearch(arr,i,elm);
  if(r==-1)
  printf("Element not found");
  else
  printf("Element in found at %d
position",r);
  return 0;
}
int Lsearch(int*a,int s,int e)
{
  int i;
  for(i=0;i<=9;i++)
  {
     if(a[i]==e)
     break;
  }
  if(i==s)
  {
     return -1;
  }
  else
  return i;
}
```

Output

```
/tmp/J7Gy8tro89.o
Enter the Array elements:- 1 2 3 4 5 6 6 7 8 9
Enter the element to be searched:2
Element in found at 1 position
```

7. Binary Search for Finding Element in an Array.

```
//Binary Search program.
#include<stdio.h>
int main(){
   int arr(int*,int,int);
   int i,a,b;
   int aray[10];
   printf("enter the array: ");
   for(i=0;i<=9;i++)
   {
      scanf("%d",&aray[i]);
   }
   printf("enter the number to find:
");
   scanf("%d",&a);
   b=arr(aray, i, a);
   if(b>=0)
   {
      printf("the number is at %d
position",b);
   }
   else
   {
      printf("element not found");
   }
   return 0;
}
```

```
int arr(int*c, int j, int d)
{
   int lower=0,upper=j-1,mid;
   while(lower<=upper)
   {
      mid=(lower+upper)/2;
      if(c[mid]==d)
      {
         return mid;
      }
      if(c[mid]>d)
      {
         upper=mid-1;
      }
      else
      {
         lower=mid+1;
      }
   }
   return -1;
}
```

Output

```
/tmp/0a0Tll5a3L.o
enter the array: 1 2 4 5 6 7 8 9 8 4
enter the number to find: 3
element not found
```

Dynamic Memory Allocation

```
#include<stdio.h>
#include<stdlib.h>
int main(){
	int n, Lavg, avg;
	int summ(int);
	printf("Enter the number of student: ");
	scanf("%d", &n);
	Lavg = summ(n);
	avg = Lavg/n;
	printf("Average marks are %d ", avg);
	return 0;
}

int summ(int n)
{
	int *p, i, sum=0;
	p=(int*)malloc(n*sizeof(int));
	if (p == NULL)
	{
		printf("Memory allocation unsuccessful");
		exit(0);

	}
	printf("Enter the marks of %d students: ",n);
	for(i=0;i<=n-1;i++)
```

```
        scanf("%d",(p+i));
        for(i=0;i<=n-1;i++)
        sum = sum+*(p+i);
        return sum;
}
```

Output :-

```
/tmp/iKSPF3WHGL.o
Enter the number of student: 5
Enter the marks of 5 students: 55 67 78 87 59
Average marks are 69

=== Code Execution Successful ===
```

Structure

```
#include<stdio.h>
struct book
{
	char name[12];
	float price;
	int pages;

};

int display(struct book*,int);
int main()
{
	struct book b[3];
	int i;
	for(i=0;i<=2;i++)
	{
		printf("Enter name, price & pages of book: ");
		scanf("%s",&b[i].name);
		scanf("%f",&b[i].price);
		scanf("%d",&b[i].pages);

	}
	display(b,i);
	return 0;
```

```
}
int display(struct book *bb, int n)
{
        int i;
        for (i=0;i<=n-1;i++)
        {
                printf("Book Name: %s \n",bb->name);
                printf("Book Price: %f\n",bb->price);
                printf("Book Pages: %d\n\n",bb->pages);
                bb++;
        }
}
```

Output :-

```
Enter name, price & pages of book: Dasbodh 3000 500
Enter name, price & pages of book: Ramayan 3000 600
Enter name, price & pages of book: Geeta 4000 700
Book Name: Dasbodh
Book Price: 3000.000000
Book Pages: 500

Book Name: Ramayan
Book Price: 3000.000000
Book Pages: 600

Book Name: Geeta
Book Price: 4000.000000
Book Pages: 700
```

Linked List

```
#include<stdio.h>
#include<stdlib.h>
struct node
{
	int i;
	struct node*l;
};
struct node*p;
int main()
{
	int append(struct node*,  int);
	int display (struct node*);
	int count (struct node*);
	int cnt;
	p = NULL;
	append(p,11);
	append(p,77);
	append(p,63);
	display(p);
	cnt = count(p);
	printf("%d", cnt);
	return 0;
}
int append(struct node *q, int n)
```

```
{
        if(q == NULL)
        {
                p=(struct node*)malloc(sizeof(struct node));
                p-> i = n;
                p-> l = NULL;
        }
        else
        {
                while(q->l != NULL)
                {
                        q=q->l;
                }
                q->l=(struct node*)malloc(sizeof (struct node));
                q->l->i = n;
                q->l->l = NULL;
        }
}
int display(struct node*q)
{
        while(q)
        {
                printf("%d\n", q->i);
                q=q->l;
        }
}
```

```
int count(struct node*q)
{
        int c=0;
        while (q)
        {
                c++;
                q=q->l;
        }
        return c;
}
```

Output :-

```
11
77
63
3
```